

MU-HUA (MAURICE) WANG

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RESEARCH INTEREST

Physical assistive robotics for long-term care. I want to build robots that carry out contact-rich care tasks on a bedridden human body — starting with diaper changing, a task that demands three things at once: perceiving a body that will not cooperate, predicting how that body moves when it is touched, and manipulating deformable material safely against skin. My work so far covers all three. I have built and clinically validated a markerless human motion capture system, written learned dynamics models of musculoskeletal bodies, and measured whether compositional motion priors survive transfer to a robot hand. I am looking for a lab where these can be assembled into a machine that does the task.

EDUCATION

National Yang Ming Chiao Tung University (NYCU) — Taipei, Taiwan

M.S., Biomedical Engineering — entered Feb 2025; on approved leave of absence since May 2025 (enrollment retained)

Took leave to move from measuring human movement to building systems that act on it. Taught full time through the leave while producing three first-author robot-learning manuscripts, and now research full time.

B.S., Biomedical Engineering — Sep 2020 – Jun 2024

- GPA 3.72 / 4.30 · Final two years 4.07 / 4.30
- **IELTS 8.0 overall** — Reading 9.0, Listening 8.5, Speaking 7.5, Writing 7.0
- Coursework: Deep Learning for Biomedical Data Analysis; Neural Engineering; Brain and Physiological Signal Analysis; Smart Medical Device Mechanical Design; Medical Device Regulatory Affairs

RESEARCH EXPERIENCE

I. Robot Learning and Embodied AI — Independent Research

Oct 2025 – present · Sole author on all three manuscripts; two submitted

A self-directed program on one question: how does an agent acquire a model of a body, and what does that model actually buy? Design, implementation, experiments, statistics and writing done independently. Every result below carries adversarial controls and multiple-comparison correction, and the negative findings are reported as findings.

Two Routes to a Motor Prior: Guidance versus a Frozen Body Model — *submitted, TMLR*

- Put two competing motor priors under one SAC backbone on MyoSuite musculoskeletal bodies: a frozen forward model learned from task-agnostic, reward-free motor babbling and used for short-horizon Dyna rollouts, against a frozen teacher policy acting through a distillation loss.
- Gave each arm the adversarial control the other had — a random-initialized teacher, a random-initialized prior — so neither win could be read as an artifact of extra parameters.
- Guidance buys early speed (50.6 mrad on myoElbow, 11.7 mm on myoFinger), but its lead decays and reverses inside the training budget on one body, while the babble-trained body model beats a random-initialized one on both.
- $n = 12$ matched seeds, paired- t 95% intervals, Holm–Bonferroni across six explicitly stated family definitions, with the cells that fail some definitions reported as failing.

Which Hand-Motion Composition Axes Are Worth Measuring — *submitted, IEEE RA-L*

- Modular motion priors assume separately learned primitives recombine and that novel recombinations stay usable. That assumption is almost never measured. I built the measurement.
- A paired split that holds out compositions and scores a naive arm against an informed one, over six axes across three public hand datasets, plus a raw-data interaction statistic that runs in minutes and predicts *whether* compositional difficulty exists.
- Reported against my own result: the statistic diagnoses presence well and magnitude badly (r falls 0.843 $\rightarrow$ 0.463 once the axis it recommended is included); matching the two arms on realized per-frame information removes most of the modular advantage; and carried onto a Shadow Hand, 16 of 17 defined comparisons fail to separate novel compositions from familiar ones.

What Master-Then-Reorganise Does Not Buy: An Invariance Argument for Body Codes

- Developmental robotics proposes mastering a competence on one body then reorganising it for a new one. I show an identity settles a large family of those proposals in advance: with the dynamics core frozen and only a low-dimensional body code θ adapted, replacing θ by $M\theta$ for any invertible M and the code's weight block by $W\theta M^{-1}$ leaves the function unchanged, so adaptation begun at $M\theta_0$ under an equivariant rule cannot change held-out error.
- Then measured the two conditions the identity needs. Linearity: function-preserving *nonlinear* re-encodings cost +0.653 and +0.875 mm ($n = 20$, $p \leq 4 \times 10^{-4}$) while linear ones are bounded above by 0.020 mm. Optimiser equivariance: under Adam even a rotation displaces error cell by cell.
- Scoped the null explicitly to open-loop prediction, and reported two attempts that failed their own pre-registered checks.

II. RGB Multi-View 3D Markerless Motion Capture — NYCU

Research Assistant, Oct 2023 – Sep 2024 · Team Lead, Oct 2024 – Sep 2025 Advisor: Prof. Hui-Ting Shih

Real-time whole-body pose sensing for clinical use — the perception layer a robot working over a bed has to have. I wrote the pipeline, ran the validation experiments, and collected the clinical data myself.

- **Built the real-time pipeline end to end.** GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT triangulation with occlusion handling; and **ROWV**, an outlier-rejection algorithm for small-sample settings, faster and more accurate than KNN and RANSAC on this problem.
- **Ran the validation experiments myself.** VICON comparison over 21 participants, plus a study over 104 walking trials from 11 participants on a nine-camera ring showing that per-joint verdicts between triangulation variants depend on the joint-centre offset they contain.
- **Ran the field data collection.** 200+ stroke and knee-pain patients (Dr. Ho, Vietnam); 40+ stroke patients with exoskeleton (Kaohsiung Chang Gung Memorial Hospital); 20+ elderly participants (Kaohsiung daycare centre); real-time feedback trials with lower-limb amputees (NYCU Physical Therapy and Assistive Technology). Hospital wards, clinics and daycare centres, not a laboratory.
- **Asked how much sensing error the downstream decision can absorb** — forward-dynamic simulation with a Geyer–Herr reflex controller over 66 design cells, showing identification of a weak muscle group falls from 0.845 to 0.549 as joint-angle error grows $0^\circ \rightarrow 12^\circ$, and that mild weakness is never identified reliably at any error level.
- **Shipped the whole product**, not only the algorithm: clinician-facing GUI (PyQt6), SQL and cloud pipeline with ICD-10 coding for longitudinal and cross-sectional search, and an Unreal Engine rehabilitation game.
- **Led a 5-person R&D team** in the second year and drove the system into clinical deployment: 4 conference presentations and 6 media features (CNA, PTS News, Yahoo! News Taiwan, iThome, OCAC, NYCU).
- **Ran the collaborations**, across institutions that do not normally open up to a student: 2 signed MOUs (Taipei Veterans General Hospital; Taipei City Hospital), Kaohsiung Chang Gung Memorial Hospital, the Institute for Information Industry (cloud patient database with ICD-10 coding), an international site in Vietnam, a community daycare centre and a private clinic in Kaohsiung, a sports-performance company, and NYCU Physical Therapy and Assistive Technology. Getting a ward, a clinic and a daycare centre to say yes is a large part of why the data exists.

III. Circular-Geometry Representation of Behavioural Trajectories — Taoyuan General Hospital

Jan 2023 – present · Advisor: Dr. Tso-Jeng Wang

A methods problem disguised as a clinical one: the discriminating feature of a dose trajectory is its *shape*, but shape is confounded with magnitude and lives on a space that Euclidean clustering gets wrong.

- **Built the representation.** Normalised each daily consumed-dose series to its own value seven days earlier, took the extreme three-day moving-window slopes on each side of the test, and mapped the (post, pre) slope pair to a polar angle $\theta = \text{atan2}(\text{post}, \text{pre})$. Shape becomes comparable independent of absolute magnitude, and the feature space becomes a torus.
- **Clustered on the manifold, not in the plane.** Fitted a bivariate (product) von Mises mixture by EM on the circle, because Euclidean clustering does not respect wrap-around at $0^\circ/360^\circ$. For the stable-dose branch, reduced 114 variables by PCA and cut a circular kernel-density estimate at its valleys. Nine patterns emerged, spanning positive/negative occurrence ratios 1.64 to 0.50.
- **Attacked my own result.** Outcome-blind re-clustering, 500 bootstrap refits per branch, within-patient conditional logistic regression, restriction to windows containing no interpolated day, and reporting across $K = 4\text{--}8$ because K was an investigator choice and no criterion selected it.

- **Reported what did not survive.** Post-test escalation magnitude (OR 1.36 → 1.06 within patients) is not claimed; the stable-dose branch is labelled exploratory; the association did not replicate across a temporal split; and the trajectory adds essentially nothing predictive over attendance. Pre-test escalation did survive, and strengthened (1.15 → 1.26 per SD, $p = 3 \times 10^{-7}$).
- Scale: 9,900 analysable urine tests from 552 patients over 2018–2022, from a 20,108-record extract; earlier phase covered 1,000+ patients from the Taiwan National Health Insurance Database with nonlinear transformation, PCA and an LSTM validation model.

IV. EEG and EMG Sleep–Epilepsy Semi-Automated Analysis — NYCU

Oct 2022 – Oct 2023 · Advisor: Prof. You-Yin Chen

- Built a signal pipeline processing 24-hour recordings in 3 minutes (detrending, downsampling, filter design, time–frequency conversion, outlier detection) with automatic WAKE / NREM / REM staging from band-specific energy features, plus a manual-scoring GUI for cross-labeler consistency.
- Designed a surgical modification that raised EMG signal-to-noise ratio at the source. Collaboration with Hualien Tzu Chi Hospital and Taipei Medical University.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed conference papers

1. **Wang, M.-H.**, Tsai, K.-T., Chen, B.-H., Huang, L.-W., Tsai, C.-H., Hsu, C.-S., & Chen, Y.-Y. Markerless Gait Analysis for Lower Back Pain: A Cost-Effective Solution to Assess Pelvis–Trunk Coordination. *SEMBA 2025*.* (first author)
2. Huang, L.-W., **Wang, M.-H.**, Chen, B.-H., Su, K.-H., Tsai, C.-H., Chen, Y.-Y., & Shih, H.-T. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users Using a Markerless Motion Capture System*. **ASB 2025**.
3. Huang, A.-Y., **Wang, M.-H.**, Huang, L.-W., Chen, B.-H., Wang, H.-H., Kuo, C.-H., Lu, S.-W., & Lo, Y.-C. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle*. **SEMBA 2025**.
4. Chang, C.-W., **Wang, M.-H.**, Lin, Y.-C., Wang, C.-Y., Li, S.-J., Chen, T.-C., Liang, Y.-W., Chen, C.-T., Chen, Y.-Y., & Lin, S.-H. *Investigating the Effect of Central Thalamic Deep Brain Stimulation on Sleep in an Alzheimer's Disease Model*. **ISMRM 2024**.
5. Chen, B.-H., Tsai, C.-H., Huang, L.-W., **Wang, M.-H.**, Hsu, C.-S., & Chen, Y.-Y. Robust Self-Occlusion Mitigation for Enhanced Accuracy in Multi-View 3D Markerless Motion Capture Systems. *ICMST 2024*.*

Submitted

R1. **Wang, M.-H.** (sole author). *Which Hand-Motion Composition Axes Are Worth Measuring*. **IEEE Robotics and Automation Letters**, submitted Sept 2026. R2. **Wang, M.-H.** (sole author). *Two Routes to a Motor Prior: Guidance Versus a Frozen Body Model*. **Transactions on Machine Learning Research**, submitted Sept 2026.

Manuscript drafts — available on request

Robot learning

M1. **Wang, M.-H.** (sole author). *What Master-Then-Reorganise Does Not Buy: An Invariance Argument for Body Codes and Its Two Boundaries*.

Human motion sensing and clinical biomechanics

- M2. **Wang, M.-H.**, & Shih, H.-T. *How Much Joint-Angle Error Can the Identification of a Weak Muscle Group Tolerate? A Predictive-Simulation Study*.
- M3. **Wang, M.-H.**, & Shih, H.-T. *Joint-Centre Offset Dominates 3D Position Error and Can Reverse Comparisons Between Markerless Triangulation Methods*. (targeting *Journal of Biomechanics*)
- M4. **Wang, M.-H.**, & Shih, H.-T. *Distinguishing Graded Plantarflexor Hyperreflexia from Graded Dorsiflexor Weakness in Simulated Post-Stroke Gait: A Search over 81 Sagittal Kinematic Variables*.
- M5. **Wang, M.-H.**, Chen, B.-H., Huang, L.-W., Chen, Y.-Y., & Shih, H.-T. *Validation of the NTKCAP Markerless Motion Capture System in Gait Kinematics*.

Behavioural data science

M6. **Wang, M.-H.**, & Wang, T.-J. *Consumed Methadone Dose Patterns Around Randomly Scheduled Urine Testing: Pre-Test Escalation and Higher-Risk Trajectories in Long-Retained Maintenance Patients*.

TEACHING

Physical Science Teacher (Grade 8) — Tamkang High School, New Taipei · Feb 2026 – Aug 2026

Taught the full Grade-8 physical science curriculum to two of the grade's eight classes.

- **Class 8 finished 2nd of eight classes on the third term exam, up from 3rd**, with its mean margin over the school average widening from +5.1 to +11.8 points across three term exams.
- **The gain came from the bottom of the distribution, not the top.** By the third exam, Class 8's bottom-quartile cutoff (78) had reached the school-wide median, and its bottom-12% cutoff stood 28 points above the school's — up from a 14-point gap on the first exam.
- Class 8 held the second-narrowest score spread in the grade (SD 15.07 against a school-wide 21.86): the class improved by pulling up students who were behind, not by stretching students already ahead.
- **Designed the mechanism that produced it.** The class earned its reward only if *every* student scored 100 — a 10-minute paper, then open discussion, then a fresh paper, repeated until the whole class cleared it. The collective condition put the strongest students to work teaching the weakest, which is where the measured gain came from.
- **Ran self-initiated weekly remedial sessions** outside class hours across three grades, targeted at the students who were behind — not assigned by the school.
- Ran hands-on chemistry and physics activities outside the syllabus, and class events to build cohesion.

HONORS AND AWARDS

2nd Place — Youth AI Startup Collaborative Hackathon, Ministry of Digital Affairs · Fall 2023 · Taipei

Led a team building the business model and commercialization path for a markerless visible-light motion capture system. Prize US\$1,000.

1st Place — Competition for Innovative Proposals in Elderly Leisure, Health and Technology Practices, National Chi Nan University · Fall 2023 · Nantou

Proposed a fall-detection and emotional-gait-recognition model for elderly care. Prize US\$340.

TECHNICAL SKILLS

Languages	Python (advanced), MATLAB (advanced), C/C++ (intermediate)
ML / RL	PyTorch, TensorFlow; SAC and Dyna-style model-based RL; policy distillation; world models; CNN/VGG; LSTM
Robotics & simulation	MyoSuite; musculoskeletal forward dynamics; Geyer–Herr reflex control; Shadow Hand; inverse kinematics; DLT triangulation; Unreal Engine
High-performance computing	CuPy, Numba, multiprocessing, GPU acceleration, CPU affinity and priority tuning
Statistics & experiment design	matched-seed paired- <i>t</i> designs; Holm–Bonferroni with stated family definitions; FDR control; von Mises / directional statistics; bootstrap reproducibility; cluster-robust and conditional logistic regression; adversarial and negative controls; pre-registered checks
Systems	PyQt6, SQL, cloud data pipelines, ICD-10 clinical coding

LEADERSHIP AND ACTIVITIES

- **Leader, Rural Service Team** · Fall 2024 – Fall 2025 — led two 3-day service camps in Tamsui and Fenglin combining educational workshops with community service. **Programme lead, Beitou church service team** (2026).
- **Music Director, church worship team** · present — and **Founder & President, ToGather Music (YouTube)** · Fall 2021 — a team of 3 producers and 5 vocalists, a self-built recording studio, 13 original songs published. Five years of continuous responsibility for a performing team.
- **Long-term mentor** to a group of four students spanning primary to senior-high school.
- **Chief Organizer, Biomedical Engineering & Nursing Student Camp** · Fall 2022 — 50 students across two disciplines. **Graduation Representative** · Fall 2024.